

Analysis of the meaning of emojis in text (Group 13)

GitHub Repository

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Abstract

This work explores how textual and embedding-based representations capture the emotional semantics of emojis. The main goal here is to assess the coherence between human-annotated emotion scores and computational emotion representations that are derived from emoji textual definitions and social media usage data. We reach this aim by utilizing several different natural language processing techniques. First, more traditional word embedding models-Word2Vec, GloVe, and FastText-compute the vectors of emotion similarity, which are then compared with the human annotations of the EmoTag1200 dataset using Pearson correlation. Next, we apply contextual and lexical embeddings to large-scale tweet datasets for modeling semantic distances among emojis, represented with t-SNE. Further, topic modeling approaches like LDA and BERTopic map emoji-related tweet corpora into emotion spaces, thus facilitating data-driven emotion alignment. We also test the DeepMoji model on emoji prediction from text and carry out antonymy-based lexical analysis regarding semantic opposition between emojis. Results indicate that subword and transformer-based embeddings excel at modeling the nuances of affect compared to classical models. Among these, FastText and DistilBERT show the best alignment with human emotion labels. BERTopic results in a higher coherence measure of topics than LDA, on average with a cosine similarity of approximately 0.73, while DeepMoji performs well in the top-5 prediction accuracy for unambiguously emotional emojis, reaching 0.71. Antonymy analysis shows weak lexical opposition, pointing out limitations with regard to using symbolic methods to model the emotion of multimodal communication. In all, this work demonstrates the value of contextual embeddings and transformer-based topic modeling toward the emotion-aware analysis of text and emoji. It contributes to the field of affective computing, and by extension, sentiment analysis, integrating the linguistic and visual modalities to provide insights for more emotionally intelligent natural language processing systems.

Contents

I	Introduction	2
II	Word2Vec-based embedding compared with human annotations	2
III	Distance between emojis	4
IV	LDA-based topic modelling for emoji–emotion matching	6
V	BERTopic-based topic modelling for emoji–emotion matching	7
VI	DeepMoji-based emoji prediction from text	8
VII	Antonym-based evaluation of emoji opposition	9
VIII	Conclusion	10

I. Introduction

Emojis have become an essential component of modern digital communication, shaping the way individuals express emotions and intentions in textual interactions. Since their introduction in the late 1990s, emojis have evolved into a complex visual language capable of conveying affective and contextual nuances that plain text often fails to capture [1]. They serve as non-verbal cues that enrich computer-mediated communication by compensating for the absence of tone, facial expressions, and gestures [2]. In many cases, emojis can modify or even replace text, contributing to the emergence of a hybrid linguistic system that integrates verbal and visual elements [4].

Recent studies have investigated the semantic and emotional roles of emojis in online discourse. Wijeratne et al. [4] explored emoji meaning representation through EmojiNet, a machine-readable emoji sense inventory, while Novak et al. [5] demonstrated that emojis are powerful indicators of sentiment in social media contexts. Despite these advances, the mechanisms through which emojis embody emotional meaning and interact with linguistic content remain insufficiently understood. Capturing the emotional coherence between textual expressions and emoji usage poses a significant challenge, particularly given the variability of emoji interpretation across cultural and contextual boundaries [6].

Traditional word embedding models, such as Word2Vec [7], GloVe [8], and FastText [9], have been widely used to capture distributional semantics in textual data. However, these models often struggle with polysemy and lack sensitivity to context, which limits their effectiveness in modeling emotion-rich communication. In contrast, transformer-based models such as BERT [10] offer context-dependent representations that better reflect the subtleties of language use, but their capacity to model multimodal emotional semantics, including emojis, remains underexplored.

The goal of this project is to systematically evaluate the coherence between annotated emotion scores and textual or emoji-based definitions using multiple embedding strategies. By combining embedding models with topic modeling and clustering techniques, the study aims to visualize and interpret emotional proximities, assess the representation of antagonistic emotional pairs (e.g., joy–sadness, trust–disgust), and predict emoji usage in given textual contexts.

The novelty of this work lies in its comparative and interpretative approach to emotion modeling, integrating traditional and transformer-based embeddings to understand how emotions are distributed across linguistic and visual modalities. This research contributes to the broader field of affective computing and sentiment analysis by addressing methodological challenges in emotional representation and providing insights that may enhance the emotional sensitivity of natural language processing systems.

II. Word2Vec-based embedding compared with human annotations

We first obtain the textual definitions of the emojis from opensource datasets. Then we encoded that definition using the pre-trained Word2Vec model by averaging the embeddings of all words in the definition, producing one emoji embedding vector. We also obtain an embedding vector for each of the eight emotion labels ("joy", "sadness", "anger", "fear", "disgust", "surprise", "trust", "anticipation"). Then, for each emoji, we compute the cosine similarity between the emoji-definition embedding and each of the eight emotion embeddings. The result is an 8-dimensional similarity vector describing how semantically close this emoji is to each emotion. Finally, we compute the Pearson correlation coefficient between the 8-dimensional human annotation vector from EmoTag1200¹ and the 8-dimensional cosine-similarity vector derived from the emoji's textual definition. This correlation quantifies how well the textual definition aligns with the human emotional labelling.

a) Word2Vec embedding: Given the set of emoji definitions, we first use `re.compile` function to convert definition strings into pattern object. Then use Word2Vec model to compute the definition's embedding vector and each emotion's embedding vector.

b) Cosine similarity: Given each emoji definition embedding vector A and each emotion embedding vector B , we compute their cosine similarity as:

$$\begin{aligned} \text{sim}(\mathbf{A}, \mathbf{B}) &= \cos \theta \\ &= \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} \\ &= \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}} \end{aligned} \quad (1)$$

then we combine them to form an 8 dimension vector C as:

$$\mathbf{C} = \begin{bmatrix} \text{sim}(\mathbf{A}, \mathbf{B}_1) \\ \text{sim}(\mathbf{A}, \mathbf{B}_2) \\ \vdots \\ \text{sim}(\mathbf{A}, \mathbf{B}_8) \end{bmatrix}$$

This predicted vector is provided to compare with human annotations.

c) Compare with Pearson Correlation: Given the similarity vectors, for each $\mathbf{C} \in \mathbb{R}^8$, we compare it's matching to the matching *EmoTag1200* emotion annotation vector $\mathbf{D} \in \mathbb{R}^8$ using Pearson Correlation:

¹<https://github.com/abushoeb/EmoTag>

$$r_{CD} = \frac{\sum_{i=1}^8 (C_i - \bar{C})(D_i - \bar{D})}{\sqrt{\sum_{i=1}^8 (C_i - \bar{C})^2} \sqrt{\sum_{i=1}^8 (D_i - \bar{D})^2}}, \quad (2)$$

$$\bar{C} = \frac{1}{8} \sum_{i=1}^8 C_i, \quad \bar{D} = \frac{1}{8} \sum_{i=1}^8 D_i.$$

The result is shown in the table and graph below. Table I presents part of the result of Pearson Correlation. The whole comparison result is shown in Figure 1.

Emoji	Pearson r	p -value
🌈 rainbow	0.271190	0.515899
🌙 crescent moon	-0.296659	0.475538
🌑 new moon face	0.202063	0.631317
☀️ sun with face	0.353910	0.389747
⭐️ glowing star	0.371975	0.364211
Mean r	0.167446	
Median r	0.205082	

TABLE I: Pearson correlation results between emoji definitions and Tag1200 emotion vectors.

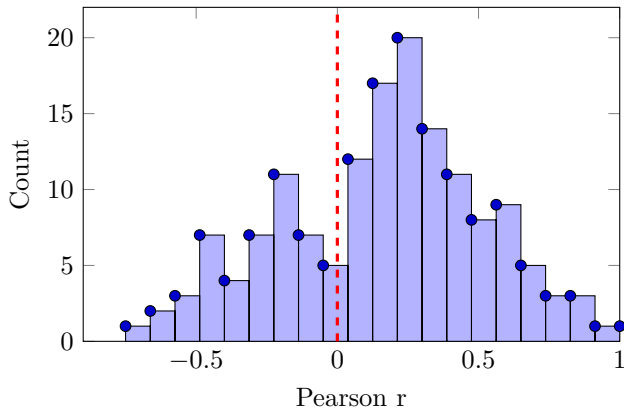


Fig. 1: Distribution of Pearson correlations across 150 emojis

The histogram shows the distribution of Pearson correlation coefficients between the model-derived 8-dimensional emotion similarity vectors and the human-annotated emotion scores from EmoTag1200 for 150 emojis. Most of the correlations fall between 0.0 and 0.5, indicating weak to moderate positive relationships. This suggests that the semantic information derived from emoji textual definitions captures some emotional tendency consistent with human annotations, but not strongly. Some emojis even show negative correlations, meaning that for some symbols, the textual definition may reflect a different or opposite emotional interpretation.

d) Other embedding models: Apart from the Word2Vec model, we also used other embedding models.

Figure 2 compares the distribution of Pearson correlation coefficients across 150 emojis for two embedding models: GloVe and FastText. The GloVe-based correlations are broadly distributed around zero, with a substantial number of negative values, suggesting that GloVe embeddings capture emotional associations only weakly and inconsistently. In contrast, the FastText model shows a clear rightward shift, with most correlations lying between 0.0 and 0.5, indicating a stronger and more consistent positive relationship between emoji textual definitions and human emotion annotations.

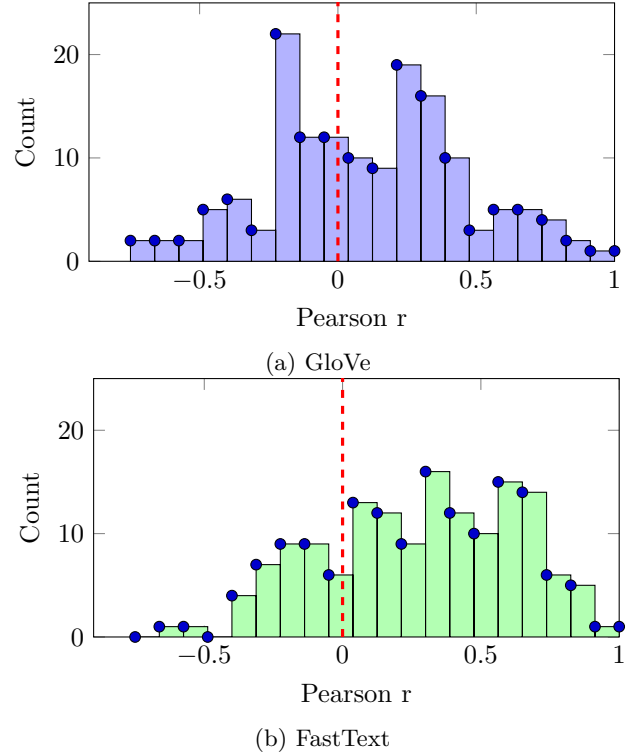


Fig. 2: Comparison of Pearson correlation distributions.

e) Discussion: Overall, the approach based on word-embedding similarity to emotion words provides a limited but non-random alignment with human emotion labeling. The FastText embeddings exhibit a more reliable alignment with human emotion annotations than GloVe, indicating that subword-based representations are more effective in modeling affective semantics of emoji definitions. Despite this improvement, the overall correlations remain modest (typically between 0.0 and 0.5), implying that textual definitions alone provide limited emotional information. Emojis are inherently multimodal and context-dependent: their perceived emotion depends on visual appearance, usage context, and cultural background. Thus, while word embeddings can approximate general affective tendencies, they cannot fully model the rich emotional semantics that humans associate with emojis.

III. Distance between emojis

III-A Distance with common words matrix

We now search a way to tell if two emojis are near each other in term of meaning and use. We will now use the Tweets with emojis dataset obtained from kaggle². It contains approximately 20,000 tweets per emojis. Stopwords have been removed from these tweets, and the words have been tokenized.

a) Objective of common words matrix: The first idea to tell if two emojis are close is to use the common words matrix between the most frequent words used in the tweets with the emojis.

b) Principle of common words matrix: To build the common word matrix, the first step is, for each emoji, calculate the first 30 most frequent words that appear in the tweets alongside each of them. Once we have this data, the next and final step is, for each pair of emojis, calculate the number of words that are present in the list of 30 most frequent words of both emojis. This gives 3.

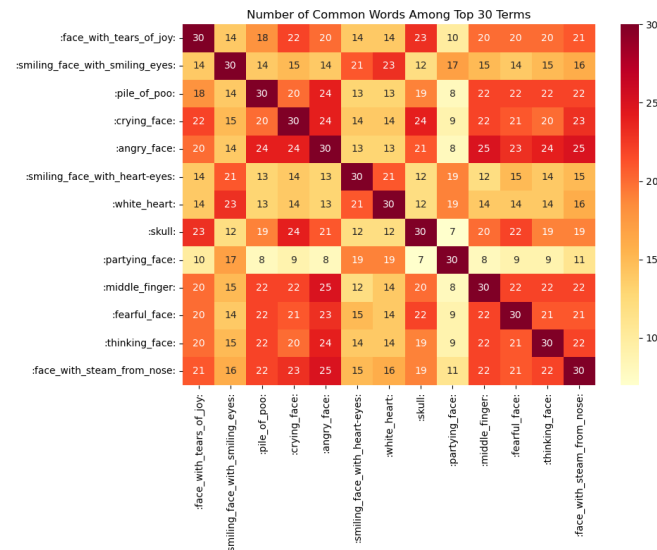


Fig. 3: Number of common words among the 30 most frequent

c) Illustrative results: To explain some of the results, according to 3, the party emoji (emoji number 9) is far from every other emojis except happy, heart eyes and white heart emojis.

III-B Distance via distilBERT embeddings and tSNE visualisation

The next idea to measure distance is to use embeddings. The first embedding used is the distilBERT embedding.

²<https://www.kaggle.com/datasets/ericwang1011/tweets-with-emoji>

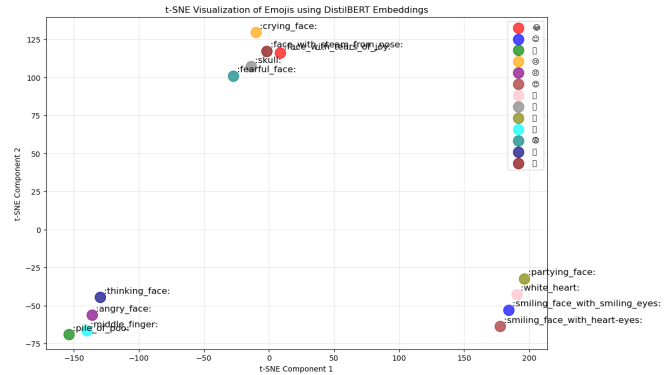


Fig. 4: t-SNE visualisation of emojis using distilBERT embedding

a) DistilBERT: is a smaller, faster, and lighter version of BERT obtained through knowledge distillation. It retains about 97% of BERT's language understanding performance while using 40% fewer parameters. DistilBERT produces contextual embeddings, capturing semantic relationships between words in a sentence. Because of its efficiency, it is widely used for embedding generation, text classification, and semantic similarity tasks.

b) DistilBERT for emojis: The problem is that embedding only exist for the words in the tweets, not emojis. Hence, the solution was to use the means of the embeddings of every tweets associated with an emojis to get the embedding of an emoji.

c) Visualisation via t-SNE: Once we have the embedding of each emojis, the last step is to visualize via t-SNE. t-SNE was chosen because it preserves local structure in high-dimensional data, making it suitable for exploring semantic neighbourhoods among emojis. However, it does not preserve global distances, meaning that only relative cluster proximity is meaningful. The random seed has been fixed for reproducibility.

d) Discussion: Quite a few things are interesting from Figure 4. First of all, it's clear that some clusters are formed, showing that some groups of emojis have similar meaning or context of use. Here, we have three clusters. One consists of positive emojis such as 🤗, 😊, 🥰, and 🥳. In another cluster, we find 🤔 mixed with more negative emojis such as 😞, 🤡, 😬 and 😫. This suggests that 🤔 may be used with ambiguity and irony. Therefore, the second cluster may be considered as an ambiguous or ironic cluster. The last cluster consists of 🤨, 🙄, and 😏. While 🤨 appearing here is somewhat unexpected, this group can be interpreted as a negative cluster.

e) Limitations: One limitation is that this approach is based on how users employ emojis in tweets and not on

TABLE II: BERT-based Distances Between Emojis

	😂	😊	💩	😭	😡	😘	❤️	💀	🔥	👉	😱	😟	😓
😂	0.00	243.75	246.17	22.97	225.16	246.85	241.31	24.23	239.44	235.35	38.94	212.01	10.57
😊	243.75	0.00	338.45	266.52	320.38	12.65	11.96	254.55	23.88	324.48	261.52	314.20	252.18
💩	246.17	338.45	0.00	245.09	21.67	331.57	345.18	224.77	352.17	14.09	211.78	34.23	240.23
😭	22.97	266.52	245.09	0.00	224.69	269.45	264.19	22.93	262.39	235.17	33.46	211.31	14.78
😡	225.16	320.38	21.67	224.69	0.00	313.91	326.76	204.06	333.43	10.75	191.29	13.48	219.43
😘	246.85	12.65	331.57	269.45	313.91	0.00	24.61	256.57	36.53	317.67	262.90	308.15	254.96
❤️	241.31	11.96	345.18	264.19	326.76	24.61	0.00	253.13	11.92	331.16	260.69	320.20	250.03
💀	24.23	254.55	224.77	22.93	204.06	256.57	253.13	0.00	252.27	214.42	14.72	190.76	15.72
🔥	239.44	23.88	352.17	262.39	333.43	36.53	11.92	252.27	0.00	338.12	260.41	326.52	248.44
👉	235.35	324.48	14.09	235.17	10.75	317.67	331.16	214.42	338.12	0.00	201.75	24.19	229.74
😱	38.94	261.52	211.78	33.46	191.29	262.90	260.69	14.72	260.41	201.75	0.00	177.92	30.14
😟	212.01	314.20	34.23	211.31	13.48	308.15	320.20	190.76	326.52	24.19	177.92	0.00	206.17
😓	10.57	252.18	240.23	14.78	219.43	254.96	250.03	15.72	248.44	229.74	30.14	206.17	0.00

their inherent semantics. Some emojis, like 😂, are highly context-dependent and may express irony or mockery.

III-C Distance via Empath embeddings and t-SNE visualisation

a) Empath: Empath is a lexical-based tool that generates interpretable text embeddings grounded in psychological and social categories. Unlike transformer-based models such as DistilBERT, Empath relies on a fixed set of 194 semantic categories (e.g., “joy”, “anger”, “violence”) derived from linguistic patterns. Each word is represented as a vector of intensities across these categories, allowing for a more interpretable yet less context-dependent representation of text.

b) Empath for emojis: As Empath does not directly embed emojis, a similar approach as before was adopted. The embedding of each emoji was obtained by averaging the Empath embeddings of all tweets in which that emoji appeared. This means that each emoji’s representation captures the dominant emotional and thematic categories associated with its usage across tweets.

c) Visualisation via t-SNE: Once the emoji embeddings were generated, they were projected into two dimensions using t-SNE for visual exploration. As with the DistilBERT visualisation, t-SNE was chosen to preserve local semantic structure and reveal clusters of emojis used in similar emotional contexts. The random seed was fixed to ensure reproducibility of the resulting layout.

d) Discussion: From the visualisation in Figure 5, clear clusters can again be observed, reflecting how emojis tend to co-occur in tweets with similar emotional or topical content. The Empath embeddings separate emojis along interpretable affective dimensions, forming a cluster of positive emojis such as ❤️, 😊, 😘, and 🥰. A second, more ambiguous cluster includes 😂, 😭, and 😱, which may be used in mixed emotional contexts, combining humor and distress. Finally, a third cluster

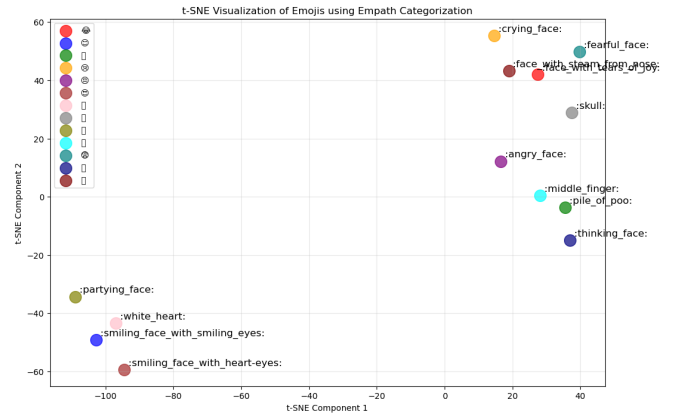


Fig. 5: t-SNE visualisation of emojis using empath embedding

groups together more negative emojis such as 😡, 👉, 💩, and 😓. Interestingly, 😟 again appears close to the negative cluster, possibly reflecting its use in sarcastic or critical contexts. The ironic and negative cluster almost form a single cluster.

e) Limitations: Because Empath relies on fixed lexical categories, it does not capture contextual nuances or irony as effectively as transformer-based models. The emoji representations therefore reflect general emotional tendencies rather than fine-grained semantic variation. Moreover, since Empath is based on English lexical norms, its performance may vary for multilingual or culturally diverse datasets.

III-D Comparison between Empath and distilBERT

a) Comparison of Representations: When comparing the two embedding approaches, several key differences emerge in how emoji semantics are captured. The DistilBERT embeddings, derived from contextualized

TABLE III: Empath-based Distances Between Emojis

	😂	😊	💩	😭	😡	😘	💖	💀	🔥	👉	😱	😟	😓
😂	0.00	158.99	46.41	18.71	31.70	158.56	150.97	16.37	156.32	41.53	14.61	57.71	8.71
😊	158.99	0.00	145.64	157.16	134.18	13.23	8.12	160.49	16.01	139.97	173.59	143.89	152.84
💩	46.41	145.64	0.00	62.82	24.86	141.43	138.36	32.73	147.72	8.57	53.75	11.30	49.94
😭	18.71	157.16	62.82	0.00	43.26	158.29	149.05	35.03	152.64	56.65	25.90	73.90	12.88
😡	31.70	134.18	24.86	43.26	0.00	132.07	126.42	26.75	133.82	16.52	44.24	34.01	31.19
😘	158.56	13.23	141.43	158.29	132.07	0.00	16.04	158.70	28.94	136.36	173.06	138.71	152.95
💖	150.97	8.12	138.36	149.05	126.42	16.04	0.00	152.64	15.13	132.53	165.57	136.91	144.77
💀	16.37	160.49	32.73	35.03	26.75	158.70	152.64	0.00	159.44	29.98	21.04	43.90	23.45
🔥	156.32	16.01	147.72	152.64	133.82	28.94	15.13	159.44	0.00	141.37	170.90	147.18	149.57
👉	41.53	139.97	8.57	56.65	16.52	136.36	132.53	29.98	141.37	0.00	50.77	17.77	43.87
😱	14.61	173.59	53.75	25.90	44.24	173.06	165.57	21.04	170.90	50.77	0.00	64.87	21.90
😟	57.71	143.89	11.30	73.90	34.01	138.71	136.91	43.90	147.18	17.77	64.87	0.00	61.03
😓	8.71	152.84	49.94	12.88	31.19	152.95	144.77	23.45	149.57	43.87	21.90	61.03	0.00

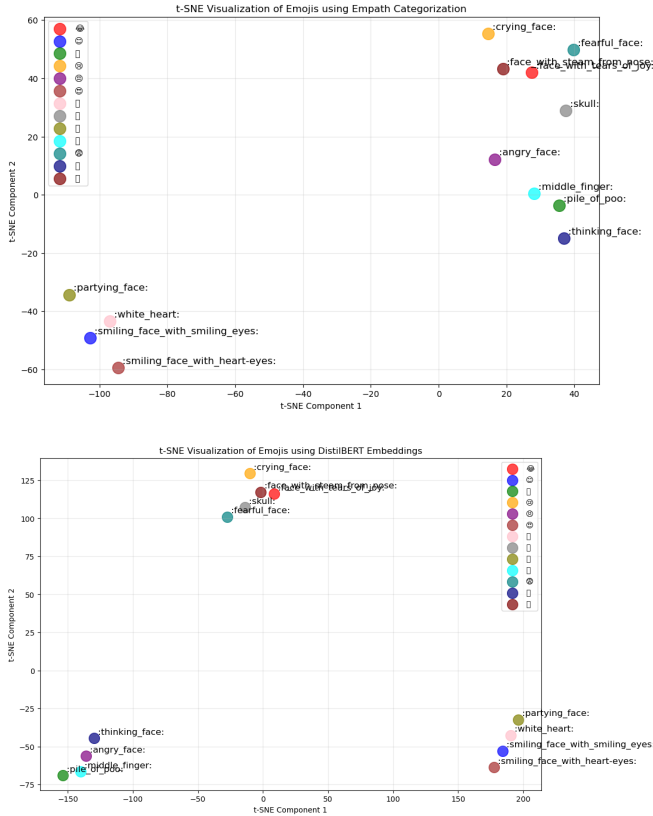


Fig. 6: t-SNE visualizations of emojis using Empath (top) and DistilBERT (bottom) embeddings.

language representations, produce more compact and well-defined clusters in the t-SNE space (Figure 6). This indicates that emojis appearing in similar linguistic and contextual environments are embedded close together, effectively capturing nuanced patterns such as irony or mixed sentiment. In contrast, the Empath embeddings (Figure 6) yield a more diffuse spatial distribution, with clusters that are less sharply delineated. This reflects

Empath’s reliance on fixed lexical categories, which limits its ability to represent subtle contextual variations in emoji usage.

b) Semantic Interpretability vs. Contextual Sensitivity: While DistilBERT offers richer and more context-sensitive embeddings, its representations are less interpretable, as the dimensions of the embedding space are not directly linked to human-understandable semantic categories. Empath, by contrast, produces embeddings that can be easily interpreted through predefined affective and topical categories (e.g., “joy”, “anger”, “sadness”). This interpretability comes at the cost of flexibility and expressiveness: Empath cannot capture sarcasm, irony, or multi-meaning emojis as effectively as DistilBERT.

c) Cluster Structure and Semantic Proximity: In the DistilBERT visualisation, emojis form three clear semantic clusters: positive, negative, and ambiguous or ironic. The ambiguous cluster, in particular, highlights how context-sensitive models can capture overlapping emotional functions—for instance, 😂 co-occurring with both positive and negative emojis due to its frequent use in humorous or ironic contexts. In contrast, the Empath embeddings reveal broader affective groupings, but with less distinction between ambiguous and negative clusters. Emojis like 😟 and 😂 appear in intermediate positions, suggesting that Empath can identify shared affective themes but not subtle pragmatic differences.

IV. LDA-based topic modelling for emoji–emotion matching

Instead of relying on predefined lexical categories such as Empath, we propose a data-driven approach based on topic modelling to explore the semantic alignment between emojis and emotions. For each emoji-labelled dataset (approximately 20,000 tweets), we apply Latent Dirichlet Allocation (LDA) using one topic and eight representative words per topic. Each tweet corpus is first

Emoji	Top 8 LDA keywords	Cosine	Pearson
🔥	one, like, love, get, easter, time, see, happy	0.90	0.37
😄	like, one, get, know, time, think, people, even	0.87	0.43
😬	like, one, people, get, know, see, would, time	0.83	0.38
👍	good, great, like, well, one, get, see, thanks	0.80	0.24
✨	happy, easter, love, like, thank, day, time, one	0.80	0.62
👻	like, one, even, bro, get, people, know, got	0.79	0.27
😱	like, one, love, good, ghost, get, day, time	0.76	0.64
🎉	happy, birthday, congratulations, day, easter, amp, good, party	0.76	0.72

TABLE IV: Examples of emojis with high LDA–EmoTag alignment.

preprocessed by removing URLs, mentions, hashtags, and non-alphabetic tokens, then lowercased and tokenized using `gensim`’s `simple_preprocess` function.

a) LDA extraction: Given the set of tweets associated with a given emoji G , we build a dictionary and a bag-of-words representation, and fit a one-topic LDA model. The model outputs a topic distribution consisting of the eight most probable keywords $\{K_1, K_2, \dots, K_8\}$ with corresponding weights $\{w_1, w_2, \dots, w_8\}$. These keywords summarize the semantic field most characteristic of the tweets containing the emoji G .

b) Mapping to emotion space: To relate these topic keywords to the emotional categories used in the *EmoTag1200* resource (*anger, anticipation, disgust, fear, joy, sadness, surprise, trust*), we compute a vector representation of the topic in the emotion space. For each keyword K_i , we evaluate its cosine similarity with each emotion name using a pre-trained embedding model (*glove-wiki-gigaword-100*). Let $\text{sim}(K_i, E_j)$ denote the similarity between keyword K_i and emotion E_j . The predicted emotion weight vector for emoji G is then defined as:

$$\text{EmojiVecLDA}(G)_j = \frac{1}{Z} \sum_{i=1}^8 w_i \text{sim}(K_i, E_j)$$

where Z is a normalization constant ensuring that the components of $\text{EmojiVec}(G)$ sum to one.

This predicted vector is compared with the ground truth emotion scores $\text{EmoTag}(G)$ provided by *EmoTag1200* using cosine similarity and Pearson correlation.

c) Illustrative results: Table IV presents examples of emojis for which the LDA keywords and emotion mappings

show high agreement with the *EmoTag* emotional profiles. Emojis not present in the *EmoTag* dataset were ignored.

d) Discussion: Overall, the cosine similarity between the predicted and reference emotion vectors is relatively high (average ≈ 0.7), showing that a single-topic LDA can recover meaningful emotional cues from tweet corpora. Keywords such as *happy, love, good, day* appear consistently for positive emojis (🔥, ✨, 🎉), while more neutral or negative emojis (😬, 👻) exhibit terms like *fear, people, even*, reflecting the semantic polarity captured by LDA. However, the method has limitations:

- common Twitter tokens (*like, get, one*) tend to dominate the topics, reducing interpretability;
- LDA, based on bag-of-words, struggles with contextual nuances (e.g., sarcasm or emoji co-occurrence);
- some emojis have no match in the *EmoTag* dataset, yielding zero-valued reference vectors.

Despite these constraints, this experiment demonstrates that topic modelling can be used to infer an interpretable mapping between emoji usage in natural text and emotion categories, forming a data-driven alternative to manually defined lexicons such as *Empath*.

V. BERTopic-based topic modelling for emoji–emotion matching

Building upon the previous LDA-based experiment, we now adopt a transformer-based topic modelling approach using **BERTopic**³. Unlike LDA, which relies on word co-occurrence patterns in a bag-of-words space, BERTopic leverages contextual sentence embeddings obtained from transformer models (here, `all-MiniLM-L6-v2`), combined

³<https://github.com/MaartenGr/BERTopic>

with clustering and class-based TF-IDF to identify semantically coherent topics.

a) BERTopic extraction: For each emoji-labelled corpus G (around 20,000 tweets), we sample up to 3,000 tweets for efficiency and apply a preprocessing pipeline identical to that used for LDA: removing URLs, mentions, hashtags, and non-alphabetic tokens, followed by tokenization and stopwords filtering using `gensim`'s `simple_preprocess` function. Each preprocessed tweet is then embedded into a dense vector using the sentence transformer model. BERTopic performs dimensionality reduction and hierarchical clustering on these embeddings to infer topics. For each emoji corpus, we select the dominant topic (the one assigned to most tweets) and extract its eight most representative keywords $\{K_1, K_2, \dots, K_8\}$ with their associated weights $\{w_1, w_2, \dots, w_8\}$.

b) Mapping to emotion space: As in the LDA experiment, we map these keywords into the eight-dimensional emotion space of *EmoTag1200* (*anger, anticipation, disgust, fear, joy, sadness, surprise, trust*). For each keyword K_i , we compute its cosine similarity with each emotion label E_j using a pre-trained word embedding model (`glove-wiki-gigaword-100`). The resulting predicted emotion vector for emoji G is given by:

$$\text{EmojiVecBert}(G)_j = \frac{1}{Z} \sum_{i=1}^8 w_i \text{sim}(K_i, E_j)$$

where Z normalizes the vector so that its components sum to one. As for LDA, the predicted vector is compared to the ground-truth emotion scores $\text{EmoTag}(G)$ through cosine similarity.

c) Illustrative results: Table V shows representative examples of emojis for which the BERTopic-derived keywords exhibit strong alignment with the corresponding EmoTag emotional profiles. Only emojis present in both datasets were retained.

d) Discussion: Overall, the cosine similarity between BERTopic-predicted and reference EmoTag emotion vectors is high (average ≈ 0.73), showing that transformer-based topic modelling captures emotional nuances more effectively than LDA. Emojis such as 🧡 and 🎉, strongly associated with enthusiasm and positivity, exhibit cosine similarities above 0.75, indicating strong semantic-emotional coherence. Meanwhile, context-dependent or ironic emojis (e.g. 🤔, 😏) show lower alignment, reflecting the inherent ambiguity of their usage.

Compared with LDA, BERTopic produces more semantically consistent and emotionally grounded keywords thanks to contextual embeddings, although high-frequency neutral terms (*like, good, know*) still appear frequently. Further refinements, such as lemmatization or filtering based on part-of-speech tags could improve topic interpretability.

Emoji	Top 8 BERTopic keywords	Cosine
🧡	love, like, time, got, good, today, let, new	0.90
😄	like, time, know, good, people, think, day, really	0.88
😮	like, know, people, going, day, said, got, god	0.82
👍	good, like, great, thank, thanks, know, time, see	0.81
💀	like, know, people, shit, think, got, time, life	0.80
🙏	thank, good, god, know, love, day, bless, like	0.77
🎉	birthday, happy, congratulations, day, thank, love, great, best	0.75
😊	like, love, happy, time, thank, day, great, good	0.71
❤️	love, day, happy, good, thank, easter, time, one	0.68
🤔	like, know, people, bull, piece, good, think, bad	0.55

TABLE V: Examples of emojis with high BERTopic-EmoTag alignment.

e) Conclusion: BERTopic provides a transformer-based extension of topic modelling that captures both contextual and emotional dimensions of emoji usage in natural text. Its superior performance over LDA in terms of topic coherence and emotion-vector similarity confirms that contextual embeddings are highly beneficial for emotion-aware semantic analysis of social media content.

VI. DeepMoji-based emoji prediction from text

In this final experiment, we reverse the direction of analysis: instead of predicting emotions from emojis, we aim to predict *emojis from text*. To do so, we use the **DeepMoji** model [?], developed by MIT and distributed by HuggingFace⁴. DeepMoji is a transformer-like neural network trained on 1.2 billion tweets containing emojis. Given an input sentence, it outputs a probability distribution over 64 commonly used emojis.

a) Emoji prediction: We use the pre-trained `torchmoji_emojis` model, which maps each input text to a 64-dimensional probability vector $\mathbf{p}$ over the emoji set $\mathcal{E} = \{E_1, E_2, \dots, E_{64}\}$. For a tweet t , the predicted emojis are obtained by selecting the k highest-probability symbols:

$$\hat{E}_k(t) = \text{Top-}k(\mathbf{p}(t))$$

where $k \in \{1, 2, 3, 4, 5\}$. This allows us to evaluate whether the ground-truth emoji G is recovered not only as the top prediction, but also among the top 2, 3, 4, or 5 candidates.

⁴<https://github.com/huggingface/torchMoji>

b) Illustrative examples: To demonstrate the behaviour of the DeepMoji model, we provide two sample predictions obtained directly from the command line interface of our script `text_emoji.py`. The model outputs the most probable emojis corresponding to the sentiment conveyed by the text.

```
$ python3 text_emoji.py --text="I am pleased
to present this research report on emoticons!"
→ I am pleased to present this research report
on emoticons! 😄👍😄😄😄😄
$ python3 text_emoji.py --text="It really
annoys me when Bert takes three hours to find
his topics!"
→ It really annoys me when Bert takes three
hours to find his topics! 😡😡😡😡😡😡
```

c) Evaluation protocol: To assess model consistency with emoji-labelled corpora, we compute, for each emoji label G (e.g., *fire*, *thumbs_up*), the proportion of tweets whose ground-truth emoji is present within the top- k predicted emojis:

$$\text{MatchProp}_k(G) = \frac{1}{|\mathcal{T}_G|} \sum_{t \in \mathcal{T}_G} \mathbb{1}_{[G \in \hat{E}_k(t)]}$$

where $\mathcal{T}_G$ denotes the set of tweets associated with emoji G . This metric thus generalizes the classical accuracy (top@1) to top- k correctness. Each emoji corpus is downsampled to at most 5,000 tweets to ensure balanced and efficient evaluation.

d) Results: Table VI summarizes the matching proportions for $k = 1$ to 5. Overall, frequently used and emotionally unambiguous emojis such as 👍 or 😄 show the highest accuracy, while context-dependent or polyvalent symbols (e.g., 😊, ❤️) remain challenging.

Emoji	Top@1	Top@2	Top@3	Top@4	Top@5
👍	0.35	0.50	0.60	0.66	0.71
😄	0.31	0.48	0.57	0.63	0.68
🙏	0.29	0.35	0.39	0.43	0.46
😂	0.19	0.23	0.25	0.27	0.28
😍	0.14	0.20	0.25	0.29	0.32
😘	0.09	0.15	0.21	0.27	0.33
🙄	0.09	0.14	0.16	0.19	0.22
👁️	0.06	0.10	0.13	0.16	0.19
👹	0.06	0.10	0.13	0.16	0.19
👹	0.05	0.09	0.14	0.19	0.26
👹	0.05	0.07	0.10	0.12	0.14
👹	0.05	0.07	0.10	0.12	0.14
👹	0.02	0.06	0.12	0.16	0.20
❤️	0.02	0.06	0.13	0.19	0.26
😁	0.02	0.04	0.06	0.09	0.12
😊	0.01	0.03	0.06	0.09	0.12
😬	0.00	0.02	0.07	0.11	0.14

TABLE VI: Proportion of correctly predicted emojis per class for top@1 to top@5 accuracy.

e) Discussion: DeepMoji achieves solid performance when the emoji expresses a clear affective intent (e.g., 👍, 🙏, 😄). As expected, increasing k substantially improves the matching rate (e.g., from 0.35 at top@1 to 0.71 at top@5 for 👍), indicating that the correct emoji is often among the top few candidates even if not ranked first. However, performance remains weaker for context-dependent emojis such as 😊 or ❤️, which frequently encode nuances like irony, affection, or politeness that text alone cannot fully capture. Moreover, DeepMoji’s vocabulary is restricted to 64 emojis, which limits overlap with the larger set used in the EmoTag dataset.

f) Conclusion: This evaluation shows that DeepMoji captures the general emotional orientation of texts with reasonable accuracy, but fine-grained emoji prediction remains difficult. Using top- k evaluation highlights that while the top prediction may often differ, the correct emoji typically appears within the top five candidates—confirming that the model encodes rich affective semantics despite noise and context variability.

VII. Antonym-based evaluation of emoji opposition

Finally, we explore a complementary linguistic evaluation based on **antonymy relations** between words, as a way to assess whether textual evidence associated with *antagonistic emojis* (e.g., 🙏 vs. 😄) exhibits opposite semantic fields. The underlying hypothesis is that emojis expressing opposing emotions should co-occur with lexical fields that are linked through antonym relations in lexical databases such as WordNet.

a) Methodology: For two selected emojis G_1 (e.g., sadness) and G_2 (e.g., happiness), we collect all corresponding tweets and extract the most frequent words after extensive preprocessing: URLs, mentions, hashtags, non-alphabetic tokens, and both standard and Twitter-specific stopwords are removed. To avoid noise from generic or sentiment-neutral words (*like*, *get*, *know*, etc.), we further restrict our analysis to words that have at least **two antonyms** listed in WordNet.

Let H_1 and H_2 be the top 20 filtered keywords for the two emojis. We define the antonym overlap Ω as:

$$\Omega(H_1, H_2) = \frac{|\{(w, a) \mid w \in H_1, a \in \text{Antonym}(w) \cap H_2\}|}{|H_1|}$$

This measure quantifies the proportion of terms in H_1 whose antonyms appear in H_2 , and vice versa. If the two emojis truly represent opposite affective poles, one would expect this overlap to be significantly higher than random chance.

b) Implementation: The experiment is implemented in Python using `nltk.corpus.wordnet` for antonym re-

trieval and `gensim`'s `simple_preprocess` for tokenization. The expanded stopwords list includes over 60 common and informal tokens frequently appearing in Twitter data, ensuring that only semantically meaningful words remain. To increase interpretability, only words with at least two antonyms were retained, thus filtering out non-informative or context-neutral terms.

c) Results.: After filtering, the 20 most antonym-rich words associated with each emoji are:

Sadness 🙄: ['first', 'bad', 'real', 'getting', 'show', 'take', 'made', 'whole', 'girl', 'new', 'miss', 'come', 'give', 'better', 'let', 'hard', 'best', 'went', 'end', 'little']

Happiness 😊: ['new', 'best', 'work', 'first', 'welcome', 'take', 'better', 'little', 'made', 'friend', 'come', 'lets', 'friends', 'keep', 'getting', 'sure', 'give', 'let', 'long', 'start']

The resulting antonym overlaps are summarized below:

$$\Omega(\text{Sad} \rightarrow \text{Joy}) = 0.15, \quad \Omega(\text{Joy} \rightarrow \text{Sad}) = 0.40$$

Example antonym pairs discovered include: *take* ↔ *give*, *went* ↔ *come*, *best* ↔ *bad*, *first* ↔ *end*, and *keep* ↔ *let*.

d) Discussion: Although a few semantically relevant antonym pairs emerge, the overall overlap remains low, indicating that lexical antonymy only weakly reflects the emotional opposition between emojis in natural text. Several factors explain these modest results:

- Many frequent tokens in social media lack explicit antonyms in WordNet, limiting lexical coverage.
- Antonymy captures logical opposites (e.g., *start–end*) rather than emotional contrasts (*happy–sad*), reducing its alignment with affective meaning.
- Emoji semantics often depend on multimodal or pragmatic cues (tone, irony, context), which are not captured by lexical databases.

Despite its limitations, this antonymy-based test provides an interesting sanity check: even if the overlap is small, the identified pairs confirm that opposing emojis tend to occur in partially distinct semantic environments. Moreover, the additional filtering step—keeping only words with at least two antonyms—proved crucial to eliminate meaningless high-frequency words and improve interpretability, though at the cost of reducing data coverage.

e) Conclusion: This final experiment demonstrates the difficulty of detecting emotional opposition purely through lexical antonymy in large-scale social media text. While symbolic resources such as WordNet offer interpretable relations, their linguistic sparsity limits

performance compared to distributional or transformer-based embeddings. Nevertheless, combining such lexical insights with topic or embedding-based models could help build richer, more interpretable representations of emoji–emotion semantics in future work.

VIII. Conclusion

Through this study, we have demonstrated that there is a very strong link between the use of emojis and the message conveyed by a text, particularly in terms of emotions. Emojis have profound meanings that can perfectly convey emotions or replace text. We often use them to replace lengthy text, to the point where we sometimes use only emojis. We also showed that emojis can be categorised (clustered), mainly according to the emotion they convey. This is due to the large number of emojis that exist.

However, this fact slowed down our study somewhat. Not all of the datasets used contained the same emojis, forcing us to reduce our panel when we had to compare two datasets.

Then, in addition to working on the meaning of emotions within the text, we also studied the opposite meaning, predicting these emojis from a text. The results were more than satisfactory, with predictions consistent with what humans might have associated with the texts. Once again, this demonstrates a strong link between text and emojis, as if we could interchange their use without affecting the meaning of the message.

Furthermore, we have identified certain limitations to the accurate understanding of the analysis of the link between emojis and text. First of all, as is well known, emojis are sometimes used ironically. This complicates their interpretation by our embedding tools (even when context is taken into account, as is the case with BertTopic).

Also, during tokenisation, we noticed that messaging-related terms (e.g., tweets) recurred frequently, which forced us to strengthen the preprocessing with an expanded list of stopwords. However, some undesirable terms inevitably remained, which certainly affected our results.

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